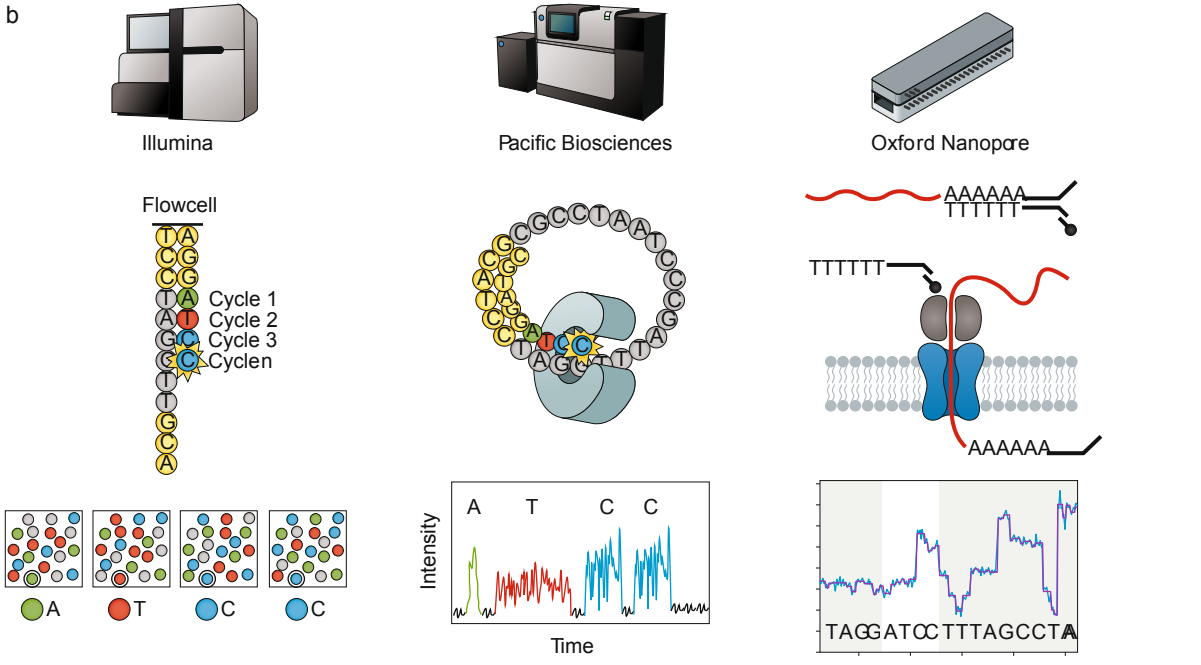
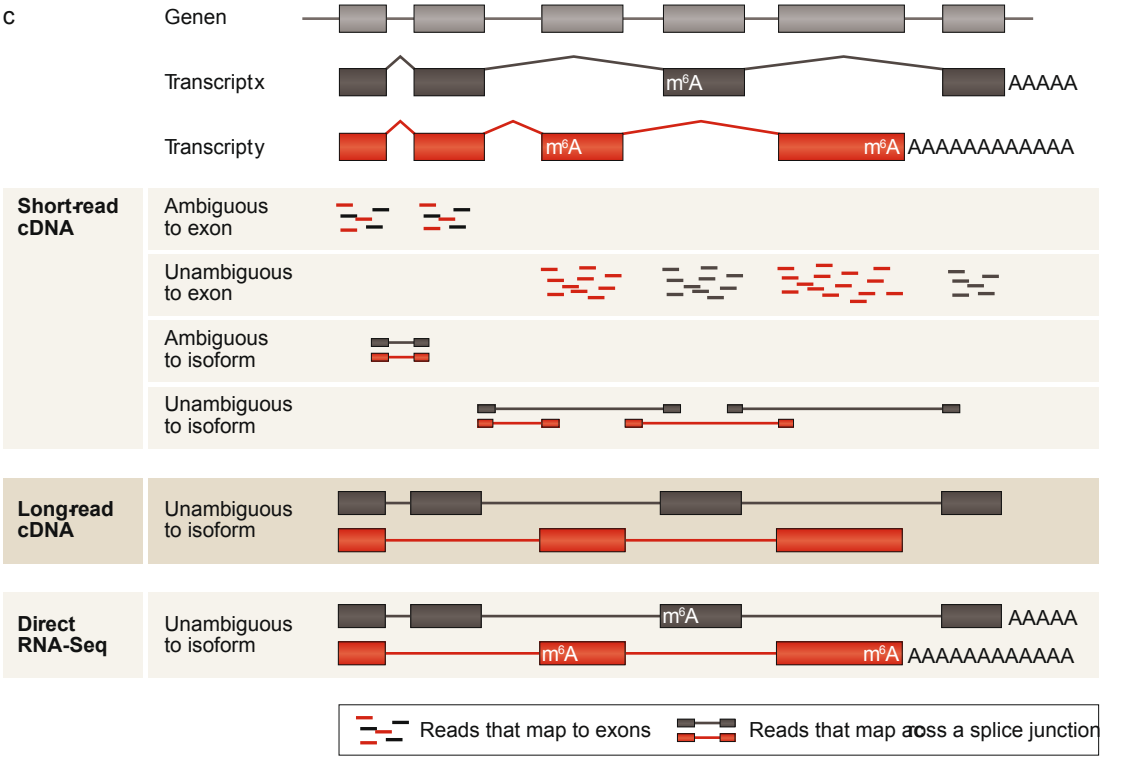

a
RNA
Direct
Longread
Shortread
Fragment RNA
cDNA synthesis
Adaptor ligation
PCR amplification
Size selection
Sequencing


b
Illumina
Pacific Biosciences
Oxford Nanopore
Flowcell
Cycle 1
Cycle 2
Cycle 3
Cyclen
AAAAAA
TTTTTT
TTTTTT
AAAAAA
A
T
C
C
Intensity
Time
A T C C
TAGGATCCTTTAGCCTA


c
Genen
Transcriptx
m6A
AAAAA
Transcripty
m6A
m6A
AAAAAAAAAAA
Short-read cDNA
Ambiguous to exon
Unambiguous to exon
Ambiguous to isoform
Unambiguous to isoform
Long-read cDNA
Unambiguous to isoform
Direct RNA-Seq
Unambiguous to isoform
m6A
AAAAA
m6A
m6A
AAAAAAAAAAA
Reads that map to exons
Reads that map across a splice junction